Do Linear Problems have Linear Optimal Algorithms

Author(s): Edward W. Packel

Source: *SIAM Review*, Sep., 1988, Vol. 30, No. 3 (Sep., 1988), pp. 388-403

Published by: Society for Industrial and Applied Mathematics

Stable URL: https://www.jstor.org/stable/2030698

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at https://about.jstor.org/terms

Society for Industrial and Applied Mathematics is collaborating with JSTOR to digitize, preserve and extend access to *SIAM Review*

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

SIAM REVIEW
Vol. 30, No. 3, September 1988

© 1988 Society for Industrial and Applied Mathematics
002

DO LINEAR PROBLEMS HAVE LINEAR OPTIMAL ALGORITHMS?*

EDWARD W. PACKEL†

Abstract. A selective introduction to the field of information-based complexity is presented in the context of the question raised in the title. After introducing some of the basic ideas of this relatively new area that interfaces mathematics and theoretical computer science, the article surveys results on the existence of linear optimal algorithms for linear problems. Several illustrative examples are given along with an indication of the variety of disciplines in which techniques of information-based complexity are being applied.

Key words. optimal algorithms, complexity, information-based complexity, linear problems

AMS(MOS) subject classifications. primary 41A45, 41A65, 65J10; secondary 46B99, 68C05

1. Introduction. The question raised in the title has both a crisp negative answer (there are counterexamples) and a surprising variety of positive ones (theorems). The general setting in which the question can be meaningfully posed involves a relatively recent research area on the interface between mathematics and theoretical computer science. This area, which we shall refer to as *information-based complexity*, is also known under aliases such as "the general theory of optimal algorithms" and "the information-centered approach to algorithms." Information-based complexity seeks to create a general theory about problems with partial or approximate information and to apply the results to specific problems. Some of the disciplines where problems with such incomplete information have arisen include computer science, economics, control theory, signal processing, and geophysics.

The goal of this article is to present a selective introduction to the field of information-based complexity and to describe the seesaw nature of results in the last decade on the extent to which linear problems have linear optimal algorithms. The approach we take will certainly not do justice to the breadth of ideas, techniques, and applications that are coming into play in the general theory. Instead, we will take a direct route to the results on linear optimal algorithms, pausing occasionally to survey the pleasing way in which ideas from linear algebra and (functional) analysis are used. In the final section we mention some other theoretical and applied topics currently under investigation in information-based complexity. A more general expository overview can be found in Traub and Woźniakowski [15] and more technical surveys appear in Packel and Woźniakowski [10] and Woźniakowski [21].

2. Two preliminary examples. We set the stage for an exposition of the general linear theory by imposing partial information on two problems which are known to virtually every student of mathematics. The problem of integrating a continuous function needs no introduction; the Fourier series approach to approximating functions by linear combinations of sines and cosines is almost as familiar. We do not develop these ideas as a testimony to the power of the general theory (more about that later), but rather to have a comfortable base to refer to when the generalizations and abstractions are introduced.

Integration example. Let f be a function from the space $\mathscr{C}[0, 1]$ of real-valued continuous functions on the interval $[0, 1]$. We would like to obtain a value for the

* Received by the editors March 3, 1986; accepted for publication (in revised form) June 5, 1987.

† Department of Mathematics and Computer Studies, Lake Forest College, Lake Forest, Illinois 60045.

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

integral of f, but, either by choice or due to computational limitations, we only have values for f at n points $t_1, \cdots, t_n$ in [0, 1]. Can we devise an algorithm to estimate the integrals of functions in $\mathscr{C}[0, 1]$ to within some predefined error ε? Some thought leads to the conclusion that regardless of how large n is, the answer is negative. Indeed, whatever function value information we may have, there are continuous functions with identical information and arbitrarily large values for their definite integrals on [0, 1]. Hence the *worst-case* error for integrals of functions in $\mathscr{C}[0, 1]$ is unbounded.

We respond in standard fashion by changing the problem so that a satisfactory estimate for the integral can be made. Define

$$F_0 = \{f \in \mathscr{C}[0, 1]: |f'(t)| \leqq 1 \text{ for all but finitely many } t \in [0, 1]\}.$$

Given a fixed n and points $t_1, \cdots, t_n$ in [0, 1], the reformulated problem is to estimate the integral of $f \in F_0$ knowing only its function values $f(t_i)$, $i = 1, \cdots, n$. As Fig. 1 indicates, the restriction on the slopes of graphs of functions now gives us bounds on functions from F_0 which have the same values as f on the points t_i. Not knowing the actual function f to be integrated, we can do no better than to integrate the function in the center of the two extremes—the piecewise linear function indicated in Fig. 1.

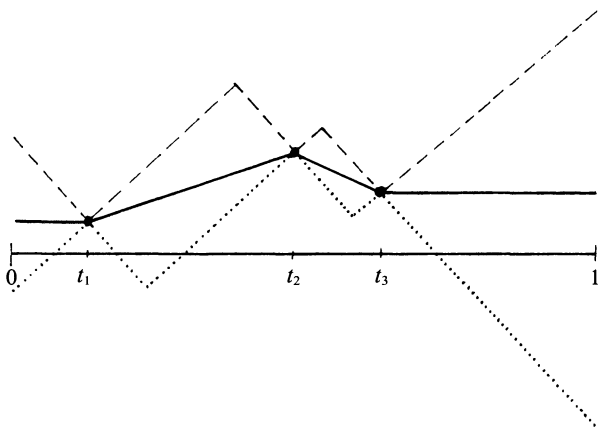


FIG. 1. *Bounds and center for functions $g \in F_0$ with $g(t_i) = f(t_i)$.*

Before we look at details, let us simplify the process by asserting that if we are free to choose the n predefined evaluation points ourselves, our best choice for minimizing the error of the definite integral estimate is to regard [0, 1] as a circle by identifying 0 and 1, and then to choose the t_i "equally spaced." Specifically, we obtain $t_i = (2i - 1)/2n$, $i = 1, \cdots, n$. In this case it can be readily checked that our best integral estimate Φ for $f \in F_0$ is precisely $\Phi = (1/n) \sum_{i=1}^{n} f(t_i)$, a particularly simple *linear* combination of the information values for f. We mention in passing that the process we have described is essentially trapezoidal integration, not the most honored of quadrature methods in terms of rate of convergence as n increases. It is, however, *optimal* as far as minimizing worst-case error (in the sense we have described) is concerned.

We also assert—and Fig. 1 tells us that this is reasonable—that the largest errors obtained by the integral estimation process described above will occur for functions f with $f(t_i) = 0$ for all i. This is illustrated for $n = 3$ in Fig. 2. It is easy to check that for such functions f, the (maximum) error resulting from our estimation algorithm is the

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

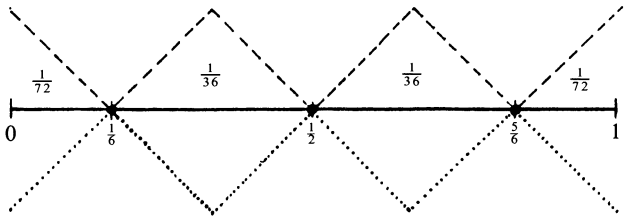


FIG. 2. *Maximum error for integral estimate with equal spacing* ($n = 3$).

area under the upper sawtooth function. The value of this worst-case error for a general n is $1/(4n)$. We conclude this example by answering the question raised at the start. Given an error tolerance ε we must have $n \geqq 1/(4\varepsilon)$ function evaluations to keep our worst case error within ε. Furthermore, *optimal information* for minimizing the error for any given n is to choose equally spaced points for function evaluation.

Fourier series example. Let $\mathbf{L}^2[0, 2\pi]$ denote the square-integrable complex-valued functions on $[0, 2\pi]$ and for $f, g \in \mathbf{L}^2[0, 2\pi]$ let the inner product $\langle f, g\rangle$ be defined by

$$\langle f, g\rangle \equiv \int_0^{2\pi} f(t)g^*(t)\,dt \quad \text{where } g^*(t) \text{ is the complex conjugate of } g(t).$$

With Lebesgue integration and some care in identifying functions which are equal "almost everywhere," $\mathbf{L}^2[0, 2\pi]$ is a Hilbert space with a norm provided by $\|f\| \equiv \sqrt{\langle f, f\rangle}$. We plan to approximate a given f as a finite linear combination of $\sin(kx)$ and $\cos(kx)$, $k = 0, 1, 2, \cdots$. To streamline matters, we choose to work with combinations of $e_k(x) \equiv (1/\sqrt{2\pi})e^{ikx}$, $k = 0, \pm 1, \pm 2, \cdots$, an equivalent framework thanks to the standard identity $e^{ikx} = \cos(kx) + i * \sin(kx)$. Once again we will impose a condition of partial information on the functions to be approximated, this time allowing $2n + 1$ inner products $\langle f, e_k\rangle$, $k = 0, \pm 1, \cdots, \pm n$, as what we know about f.

Using the Hilbert norm to measure error, we must again restrict the problem domain to achieve satisfactory results. This time we define

$$F_0 \equiv \left\{ f \in \mathbf{L}^2[0, 2\pi]: \sum_{k=-\infty}^{\infty} k^2 |\langle f, e_k\rangle|^2 \leqq 1 \right\}.$$

We are interested in algorithms $\Phi: N(F_0) \to \mathbf{L}^2[0, 2\pi]$ that approximate f in F_0 using only the information $N(f)$ about f. We wish to minimize the worst case error, $\sup\{f \in F_0: \|f - \Phi(N(f))\|\}$.

Since we have as our information on f its first $2n + 1$ Fourier coefficients, a natural algorithm Φ would be to let $\Phi(N(f))$ be the first $2n + 1$ terms of the Fourier series for f, $\Phi(N(f)) = \sum_{|k| \leqq n} \langle f, e_k\rangle e_k$. From the orthonormality of the e_k and the deeper result, which says that any f in $\mathbf{L}^2[0, 2\pi]$ "is" its full Fourier series $\sum_{k=-\infty}^{\infty} \langle f, e_k\rangle e_k$, it follows that the error for a given f is $[\sum_{|k|>n} \langle f, e_k\rangle e_k]^{1/2}$. Because of the defining condition for F_0, this error can be maximized over F_0 by choosing $f = e_{n+1}/(n + 1)^2$. By the somewhat contrived nature of the example it is not hard to see that this algorithm Φ is optimal for minimizing worst-case error, and that this worst-case error for Φ (given the $2n+1$ inner product evaluations provided by N) is precisely $1/(n+1)$, the index of the first Fourier coefficient for which we have no information.

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

Before leaving these two examples for a general development, we highlight what is common to them. In each case we sought a solution for a specified problem, the first being to obtain definite integrals (the integration operator) and the second to approximate a function (the identity operator). For any problem element f we were given only a finite number of scalars $N(f)$ as the partial information to work with; we had to restrict the problem elements to a smaller domain F_0 to get meaningful worst-case results. In both cases the solution operator and the information operator were linear mappings between normed linear spaces. Finally, we came up with the following in each case:

(i) A nonnegative real number depending on the solution operator and the information operator (and of course its cardinality) which gave the minimum possible worst-case error for solving the problem using the partial information by *any* idealized algorithm.

(ii) A particular algorithm that stayed within this minimum worst-case error, thus earning the designation of an optimal algorithm for the problem in question.

In what follows we will sometimes call upon these examples to illustrate general concepts and results. We urge the reader, as we proceed, to keep the examples in mind to whatever extent this will prove helpful as new ideas are introduced.

3. Information-based complexity for linear problems. A thorough development of the framework for information-based complexity may be found in Traub and Woźniakowski [14]. Here we present the standard approach for *linear* problems.

Let F_1 and F_2 be normed linear spaces over the scalar field K, where K is either the real numbers $\mathbb{R}$ or the complex numbers $\mathbb{C}$. The *problem elements* for which we have only partial or approximate information are assumed to belong to a convex subset F_0 of F_1. We further assume that F_0 is *balanced* ($f \in F_0$ and $\alpha \in K$ with $|\alpha| \leqq 1 \Rightarrow \alpha f \in F_0$). An important way (and in a sense the "only" way) of generating such an F_0 is to let $F_0 = \{f \in F_1 : \|T(f)\| \leqq 1\}$, where $T: F_1 \to F_4$ is a linear *restriction operator* into another normed linear space F_4. In our integration example the restriction operator was the differentiation operator (not everywhere defined) on $\mathscr{C}[0, 1]$.

A *linear problem* is defined by specifying F_0 along with two linear operators on F_1. The underlying problem solution (for which we seek approximating algorithms) is defined by a linear *solution operator* $S: F_1 \to F_2$. As we have seen, an algorithm to approximate $S(f)$ may not have full information about f to work with. Accordingly, we specify a linear *information operator* $N: F_1 \to K^n$, so that $N(f)$ gives us n (a positive integer) scalar pieces of information about the problem element f. The requirement that N has finite-dimensional range is a concession to the reality that practical algorithms process only a finite amount of information. We will often be concerned only with the behavior of S and N on the convex and balanced subset F_0 of F_1. In this case, we regard an operator defined on F_0 as linear if it is the restriction of a linear operator defined on F_1.

Recalling the two examples of the previous section, we note that the solution operators (integration and identity) were indeed linear as were the information operators (function evaluation and inner products). The first example had a particularly simple codomain F_2, namely $\mathbb{R}$, while the second example used the complex numbers $\mathbb{C}$ as its scalar field and the Hilbert space $\mathbf{L}^2[0, 2\pi]$ as its codomain.

A concept central to the whole theory is the *radius of information*, which gives the inherent error in approximating the solution $S(f)$ given only the information provided by $N(f)$. We shall work throughout this paper in the worst case error setting,

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

where the radius of information, $r(S, N)$, can be defined as follows:

(i) Given $f \in F_0$, set $V(f) = \{g \in F_0: N(g) = N(f)\}$. When we are concentrating on an information value y with $y = N(f)$, we will use the fact that $V(f) = N^{-1}(y) \cap F_0$.

(ii) Now define $r(S, N, f)$ to be the radius of $S(V(f))$ as a subset of F_2, where the radius of a subset G of a normed space F is defined by $\inf_{c \in F} \sup_{g \in G} \|c - g\|$. A point c (if it exists) for which this infimum is attained is called a *center* of G (it need not be unique).

(iii) Finally, define $r(S, N) = \sup \{r(S, N, f): f \in F_0\}$.

In the integration example, $V(f)$ is the set of functions from F_0 which agree with f on the evaluation points t_i, $i = 1, \cdots, n$. As Fig. 1 suggests, $r(S, N, f)$ is the area between either bounding sawtooth function and the piecewise linear function in the center of $V(f)$. Figure 2 concentrates on functions f that maximize $r(S, N, f)$ and the area under its upper sawtooth gives the radius of information for the problem.

In general, the key idea here is that we do not know the specific f to which the solution operator should be applied. The best we can say is that it belongs to the set $V(f)$ of problem elements which share the same information as f. Accordingly, a bound on the possible error in approximating $S(f)$ is given by the radius of the set $S(V(f))$. The radius of information for the whole problem is the "largest" such radius for f in F_0.

Closely related to this radius is the *diameter of information*, $d(S, N)$, defined by $\sup \{d(S, N, f): f \in F_0\}$, where $d(S, N, f)$ is the diameter of the set $S(V(f))$ defined above. Many results in the theory are more easily developed in terms of the diameter of information, which is not generally twice the radius, but can easily be shown to satisfy $r(S, N) \leqq d(S, N) \leqq 2r(S, N)$.

We now consider algorithms to approximate $S(f)$. Since we only have the information $N(f)$ on f, such algorithms can only be defined on $N(F_0)$. Accordingly, we define an *algorithm* Φ for the problem determined by F_0, S, and N as an operator Φ: $N(F_0) \to F_2$. In anticipation of the possible objection that our use of the term algorithm does not include a requirement such as "finitely computable," we have two responses. First, we may think of our more general notion as an "idealized" algorithm. We shall soon see that one important goal of information-based complexity is to provide a lower bound, via the radius of information, on the error of any such algorithm. Thus our more general definition of algorithm allows for stronger results, which then apply to any conventional algorithm for a given problem. Second, another goal of the general theory (as the last word of its name testifies) is to develop upper and lower bounds on the complexity of problems, so issues of computational complexity can be brought into play. We do not consider these issues further in this paper, but they are thoroughly treated in Traub and Woźniakowski [14, Chap. 5].

Of obvious importance are the *optimal algorithms*, which we now define. If we define the *error* of an algorithm Φ by

$$e(\Phi) = \sup \{\|S(f) - \Phi(N(f))\|: f \in F_0\},$$

we claim that $e(\Phi) \geqq r(S, N)$. Indeed, for any $f \in F_0$ and $g \in V(f)$, a disk with center $\Phi(N(g))$ and radius $e(\Phi)$ contains $S(g)$. Since $N(f) = N(g)$, this says that $e(\Phi) \geqq r(S, N, f)$. Taking the supremum over F_0, we get the desired inequality. An algorithm Φ^* is defined to be *optimal* if

$$e(\Phi^*) = \inf \{e(\Phi): \Phi \text{ an algorithm for } S \text{ using } N\}.$$

It follows directly that Φ^* is optimal if and only if $e(\Phi^*) = r(S, N)$. Thus $r(S, N)$ provides a tight lower bound on the error of *any* algorithm and can be viewed as the

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

intrinsic uncertainty inherent in the problem with solution S and information N. If an algorithm Φ satisfies the stronger condition that $\|S(f) - \Phi(N(f))\| \leqq r(S, N, f)$ for all $f \in F_0$, then Φ is said to be *strongly optimal* or *central*. The algorithms given in the integration and Fourier series examples both happen to be strongly optimal, but generally it is asking too much to expect realizable algorithms which are strongly optimal. We shall, however, encounter one important class of problems where this occurs.

In Fig. 3 we summarize the situation as we have described it to this point. The diagram incorporates the fact that the problem domain F_0 may arise as the inverse image of a unit ball under a restriction operator T, since we shall need this formulation at one important point.

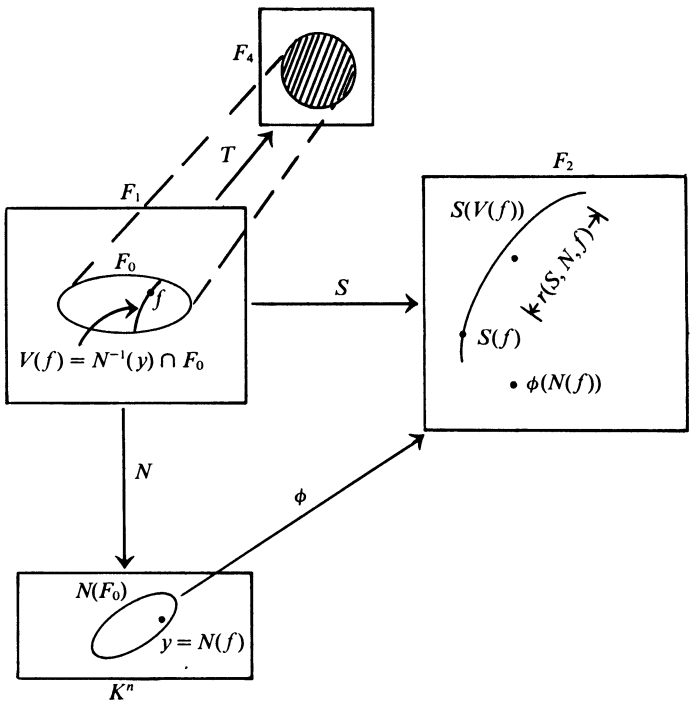


FIG. 3. *Schematic view of the general theory.*

Our restriction to problems which are linear is natural in view of the goals of this paper. While this does rule out nonlinear problems (root-finding, for example), many of the important problems for which algorithms are sought turn out to be linear. In particular, classical problems of linear differential and integral equations, integration, approximation, and interpolation are linear, as are many problems stemming from linear models in computer science and other fields.

To help fix some of the ideas above, we now review two basic results for the linear case. Recall that the kernel of a linear operator L, ker (L), is the set of domain elements mapped by L into 0.

THEOREM 1. *Given linear operators* S: $F_1 \to F_2$, N: $F_1 \to K^n$, *and* T: $F_1 \to F_4$ *with* $F_0 = \{f \in F_1 : \| T(f)\| \leqq 1\}$, *then*

(a) $r(S, N) < \infty \Rightarrow \ker(N) \cap \ker(T) \subseteq \ker(S)$;

(b) $d(S, N) = 2[\sup\{\|S(h)\| : h \in \ker(N) \cap F_0\}]$.

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

Proof. (a) Suppose, on the contrary, there exists $h \in \ker(N) \cap \ker(T)$ with $S(h) \neq 0$. Then $\alpha h \in V(0) = \ker(N) \cap F_0$ for all $\alpha \in K$. It follows that $S(V(0))$ is unbounded in F_2 and hence has infinite radius, so $r(S, T) = \infty$. This contradiction establishes the result.

(b) Given any $f \in F_0$, and any $g_1, g_2 \in V(f)$, set $h = (g_1 - g_2)/2$. Then $\|T(h)\| \leqq 1$ and $h \in \ker(N)$, so $h \in V(0)$. Thus

$$\|S(g_1) - S(g_2)\| = 2\,\|S(h)\| \leqq 2[\sup\{\|S(h)\| : h \in \ker(N) \cap F_0\}].$$

Taking the supremum first over g_1 and g_2 in $V(f)$ and then over $f \in F_0$, we get $d(S, N) \leqq 2[\sup\{\|S(h)\| : h \in \ker(N) \cap F_0\}]$. The reverse inequality follows by noting that for any $h \in V(0)$ we also have $-h \in V(0)$ (F_0 is balanced). Thus $2\,\|S(h)\| = \|S(h) - S(-h)\| \leqq d(S, N)$ and, since $h \in V(0)$ was arbitrary, the proof is complete.

Since problems with an infinite radius of information are of no great interest to us (there will always be infinite error, yet every algorithm is optimal!), we henceforth assume $r(S, N) < \infty$. As a consequence of the above theorem, we will then always have $\ker(N) \cap \ker(T) \subseteq \ker(S)$. We also note that part (b) gives partial justification for going from Fig. 1 to Fig. 2 in the integration example—we looked at the kernel of the function evaluation information operator to get the worst-case error.

For many of the same theoretical reasons that we concentrate on linear problems, we now restrict ourselves to the study of algorithms which are also linear. Such algorithms are also desirable for a variety of practical reasons, which we briefly summarize. Linear algorithms appear to be natural for problems in a linear setting. Indeed, many of the standard algorithms for classical numerical problems (integration and interpolation, for example) are linear. Linear algorithms tend to be simpler and easier to implement. Most important, linear algorithms have small combinatorial complexity and optimal linear algorithms can be formally shown to have *nearly optimal combinatorial complexity.* In addition to this valuable efficiency in time, linear algorithms also have small space complexity (if we ignore precomputations). Details can be found in Traub and Woźniakowski [14, Chap. 5]. Finally we note that in some important classical works in numerical analysis, linearity has been an assumption imposed on a class of algorithms being studied. In the light of results to be presented on the existence of linear optimal algorithms, it turns out that this assumption is not necessary. Indeed, as far as optimality is concerned, nonlinear algorithms do not help to reduce the error in these and many other important cases.

4. A more subtle example. It is easily checked that the optimal algorithms for our two preliminary examples are linear. We now develop a somewhat more obscure but, we hope, enlightening example. This example will also serve as a starting point for a subsequent counterexample.

Let both F_1 and F_2 be $\mathscr{C}[0, 1]$, continuous real-valued functions on $[0, 1]$ endowed with the supremum norm, $\|f\| = \sup_{x \in [0,1]}\{|f(x)|\}$. Let F_0 be the unit ball $\{f \in \mathscr{C}[0, 1] : \|f\| \leqq 1\}$. We take as our solution operator the indefinite integral $S: F_0 \to \mathscr{C}[0, 1]$ defined by $S(f)(x) = \int_0^x f(t)\,dt$. The information for $f \in F_0$ consists of the real number $N(f) = \int_0^1 f(t)\,dt$. Thus our problem is, essentially, to approximate antiderivatives of certain functions defined on $[0, 1]$ knowing only their definite integrals. We offer no testimonials to the relevance of this problem in any meaningful applied or theoretical setting. However, it is appropriately subtle to shed more light on the concepts of radius of information and optimal algorithm. We now proceed to calculate these, introducing in the process some geometrical ideas which will be used again later.

For each fixed $x \in [0, 1]$, consider the simpler problem with the same information N and the real-valued solution operator S_x on F_0 defined by $S_x(f) = S(f)(x)$. The

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

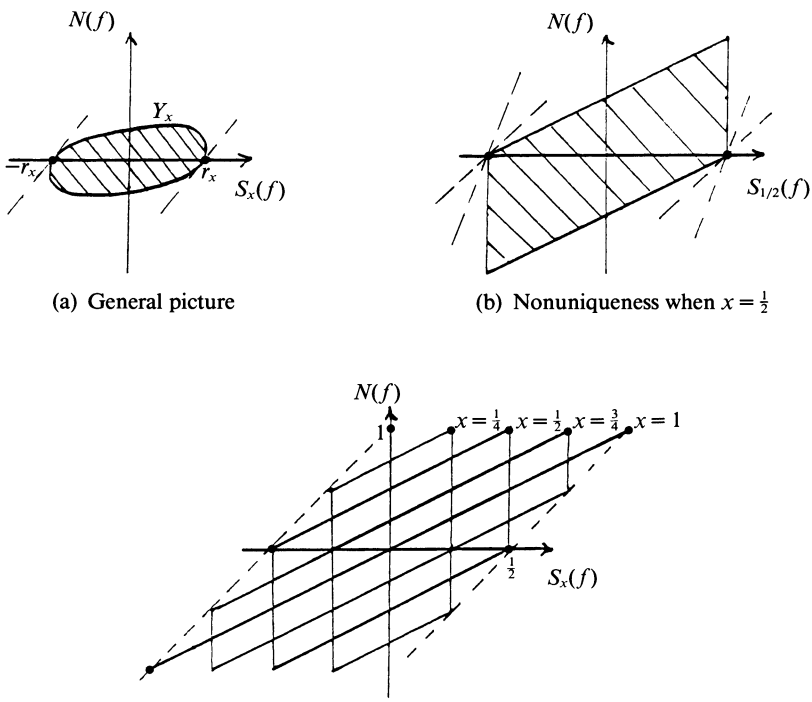
subset of the plane defined by $Y_x = \{(S_x(f), N(f)): f \in F_0\}$ is convex and balanced (this depends only on F_0 having these properties and on the linearity of the operators involved as suggested in Fig. 4(a)). There must then exist bounding parallel hyperplanes (in this case lines) tangent to Y_x at the symmetric points $(-r_x, 0)$ and $(r_x, 0)$. Such a pair of hyperplanes need not be unique (see Fig. 4(b)). This geometrical situation enables us to draw the following conclusions:

(1) For each x, $r(S_x, N) = r_x$. Indeed, the subset $S_x(V(f))$ with maximum radius occurs when $N(f) = 0$. It is the image under S_x of $\ker(N) \cap F_0$.

(2) The radius of information $r(S, N)$ for the indefinite integral S we are working with is $\frac{1}{2}$, realized as the maximum r_x value. This maximum occurs when $x = \frac{1}{2}$ (see Fig. 4(c)).

(3) For each x, a *linear* optimal algorithm for the problem consisting of S_x and N is determined by the slope of a bounding hyperplane tangent to Y_x at $(r_x, 0)$. Specifically, if the parallel hyperplanes have equations $t = \mu y \pm r_x$, then any $(S_x(f), N(f)) \in Y_x$ must satisfy $S_x(f) \leqq \mu N(f) + r_x$ and $S_x(f) \geqq \mu N(f) - r_x$. Hence $|\mu N(f) - S_x(f)| \leqq r_x$ and the linear algorithm $\Phi_x(N(f)) = \mu N(f)$ is optimal for S_x.

(4) If we now define $\Phi(N(f))(x) = \Phi_x(N(f))$, we do get a function Φ that is linear in $y = N(f)$; but $\Phi(y)$ is not (unless $y = 0$) a continuous function of x. As Fig. 4(c) indicates there must be a discontinuity at $x = \frac{1}{2}$. A linear optimal algorithm can be salvaged in this example, however, since we can find a single bounding tangent hyperplane at $(\frac{1}{2}, 0)$ (indicated by the dashed lines in Fig. 4(c)) for the *union* of all the sets Y_x, $x \in [0, 1]$. This generates a linear algorithm that gives functions which are continuous (even constant) in x. The optimal algorithm Φ^* that results is $\Phi^*(y)(x) = y/2$ for all x. In terms of earlier definitions, optimality says that $|\frac{1}{2}\int_0^1 f - \int_0^x f| \leqq \frac{1}{2}$ for all $f \in F_0$, $x \in [0, 1]$.



(a) General picture

(b) Nonuniqueness when $x = \frac{1}{2}$

(c) Y_x for $x = \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1$

FIG. 4. *Plots of* $\{S_x(f), N(f): f \in F_0\} = Y_x$.

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

We conclude that the problem we have presented does have a linear optimal algorithm, but it was a close contest. This closeness will be exploited in a later section. We now survey the brief but interesting history of the general question of when such algorithms exist for linear problems.

5. Linear optimal algorithms for linear problems. Table 1 summarizes positive and negative results on existence of linear optimal algorithms. We will discuss the items in the table essentially in chronological order. Before proceeding we mention a small bit of "prehistory." Until very recently it has been the belief of those working in the area that, despite the mounting evidence of counterexamples and the absence of a reasonably general theorem, real-world or naturally occurring linear problems *do* have linear optimal algorithms. Support for this "folk" result has been abundant, bolstered by classical numerical algorithms and some more recent applications of the work in information-based complexity. We will not supply specific examples here, but we shall see that recent developments have somewhat shaken the belief that linear problems have linear optimal algorithms.

Results 1 and 2 in Table 1 cover the important special case of problems whose solution operator and approximating algorithms require a single scalar answer for each problem element. The discussion of the example in the previous section gives a clear idea of the lovely proof of result 1. Briefly, the same tangent hyperplane (separation theorem) arguments are applied to the balanced convex set $Y = \{(S(f), N(f)): f \in F_0\}$ in $\mathbb{R}^{n+1}$. As in the example, a linear optimal algorithm emerges neatly from the inequalities generated by the tangent hyperplanes. See [14, Chap. 3] for full details.

TABLE 1
Results on existence of linear optimal algorithms for linear problems

Year reference	Summary (LOA = Linear Optimal Algorithm)
1. 1965 Smolyak [12]	LOA when $F_2 = \mathbb{R}$
2. 1976 Osipenko [7]	LOA when $F_2 = \mathbb{C}$
3. 1977 Micchelli and Rivlin [5]	LOA when $F_2 = \mathbb{R}$ and information is perturbed LOA when F_1 = Hilbert space and F_0 unit ball
4. 1978 Micchelli [3]	Counterexample with $F_1 = \mathbb{R}^3$ and $F_2 = \mathbb{R}^2$ with an l^4 norm
5. 1980 Traub and Woźniakowski [14]	LOA when $F_4 = T(F_1)$ is a Hilbert space and $T(\ker(N))$ closed in F_4
6. 1986 Packel [9]	Counterexample with $F_1 = \mathbb{R}^3$ and $F_2 = \mathbb{R}^2$ with an l^2 norm
7. 1986 Packel [9]	LOA allowing extended range for S and Φ
8. 1986 Werschulz and Woźniakowski [20]	Class of powerful counterexamples with real-world overtones

Result 3 extends result 1 to the important case where there is error or uncertainty in the information. Since the general theory is motivated by the increased realism gained in admitting limited knowledge of a problem element f through its information $N(f)$, a greater degree of realism suggests that even $N(f)$ is only known to within a certain interval of error. Even with such *perturbed* information, linear optimal algorithms exist when $F_2 = \mathbb{R}$. The second part of result 3 gives the first setting in which linear optimal algorithms must exist where the problem codomain is an arbitrary normed linear space. It is a forerunner and special case of result 5.

Result 4 makes it clear, of course, that it is too much to hope that every linear problem will have a linear optimal algorithm. Micchelli's example (which can be

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

found in [14, p. 60]) is cleverly contrived, but rather far removed from any problem that might conceivably occur naturally. Result 6 gives a slightly simpler counter-example with problem codomain $\mathbb{R}^2$ under the Euclidean norm.

Result 5 is the culmination of a series of more specialized results which link the theory of optimal algorithms with the important idea of approximation by *splines*. Our plan here is to sketch a self-contained proof of the result without relying on the theory of splines. A careful statement of result 5 now follows.

THEOREM 2. *Let $F_0 = \{f \in F_1 : \|T(f)\| \leqq 1\}$ be generated by a restriction operator T: $F_1 \to F_4$ mapping F_1 into a Hilbert space $F_4 = \mathscr{H}$ and let $T(\ker(N))$ be closed in $\mathscr{H}$. Then a linear problem determined by S and N on F_0 has a linear and strongly optimal algorithm.*

Proof. We divide the proof into 4 parts, referring to Fig. 5 and relying on a few standard Hilbert space results as needed.

(1) Given any $y = N(f) \in N(F_0)$, the subset

$$C = \{T(g) \in \mathscr{H} : g \in N^{-1}(y)\} = T(\ker(N)) + \{T(f)\}$$

is a convex and closed subset in a Hilbert space and hence has a unique element of smallest norm. Denote this element by $g(y)$. It is also the case in Hilbert space (and geometrically evident in Fig. 5) that the inner products $\langle g(y), T(h)\rangle = 0$ for all $h \in \ker(N)$ since C and $T(\ker(N))$ are "parallel" in $\mathscr{H}$. Furthermore, $g(y)$ is the unique member of C with this orthogonality property.

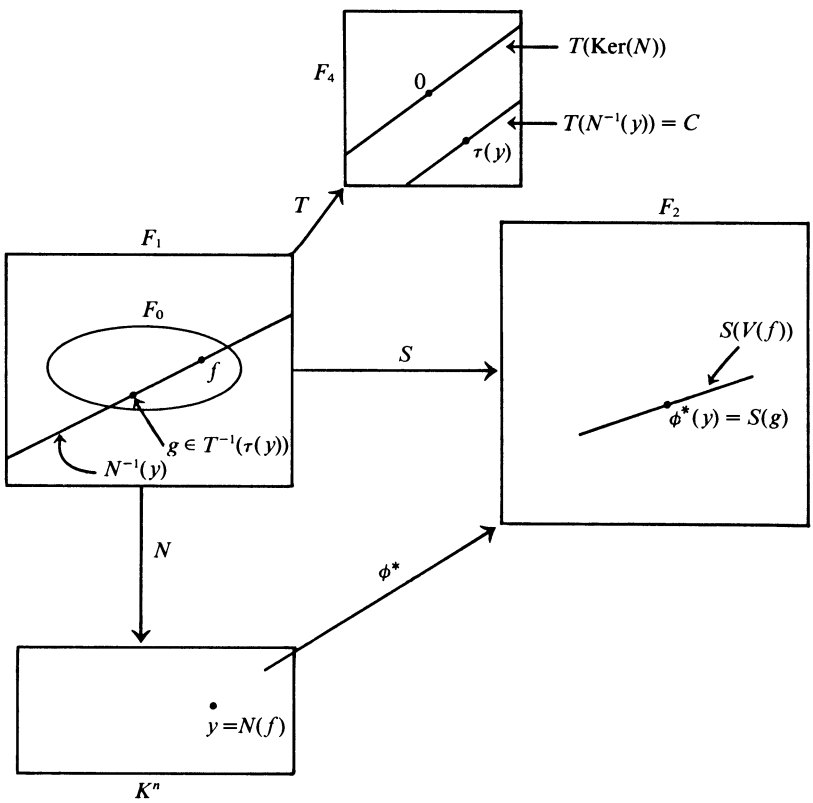


FIG. 5. *Strongly optimal spline algorithm in Hilbert space setting.*

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

(2) Let g_1, $g_2 \in N^{-1}(y)$ be such that $T(g_1) = T(g_2) = g(y)$. We show that $S(g_1) = S(g_2)$. Indeed, let $h = g_1 - g_2$ and note that $h \in \ker(N)$. Then

$$\|T(g_1)\|^2 = \|T(g_2+h)\|^2 = \|(T(g_2)+T(h)\|^2 = \|T(g_2)\|^2 + \|T(h)\|^2,$$

where the final equality comes from the orthogonality of $T(g_2)$ and $T(h)$ (the "Pythagorean theorem" generalized to Hilbert space). It follows directly that $\|T(h)\| = 0$, so $h \in \ker(N) \cap \ker(T)$. From Theorem 1(a) proved earlier we then have $h \in \ker(S)$, so $S(g_1) = S(g_2)$ as desired.

(3) We can now define $\Phi^*(y) = S(g)$, where g is *any* representative of $N^{-1}(y)$ with $T(g) = g(y)$. It is easily checked that linearity of Φ^* with respect to y results from the linearity of S and T and the previously stated orthogonality property characterizing $g(y)$.

(4) Finally, the fact that Φ^* is strongly optimal follows if we show that for each $y = N(f) \in N(F_0)$, $\Phi^*(y)$ is a center of $S(V(f))$. To this end, choose $g \in N^{-1}(y)$ with $T(g) = g(y)$ and suppose $\Phi^*(y) + k \in S(V(f))$. We can restate this (with the help of linearity) as $S(g) + S(h) \in S(V(f))$ with $h \in \ker(N)$ and we must show that, symmetrically, $S(g) - S(h) \in S(V(f))$. We may assume that $g + h \in V(f)$ (so $\|T(g+h)\| \leqq 1$), since otherwise $S(g+h) = S(g')$ for $g' \in V(f)$ and we can redefine h as $g' - g$. Orthogonality of $T(g)$ and $T(h)$ now gives $\|T(g-h)\|^2 = \|T(g+h)\|^2 \leqq 1$. Hence $g - h \in F_0 \cap N^{-1}(y) = V(f)$ and $S(g) - S(h) \in S(V(f))$. This establishes strong optimality of Φ^* and completes the proof of the theorem. We mention without proof (see [14, Chap. 4]) that Φ^* is the unique strongly optimal algorithm and the only optimal algorithm that is linear.

Result 7 resurrects, in an elegant but not totally satisfying way, the intuition that general linear problems *ought* to have linear optimal algorithms. To explain this we present a carefully stated theorem and a sketch of its proof. Fig. 6 gives a diagrammatic representation of the theorem.

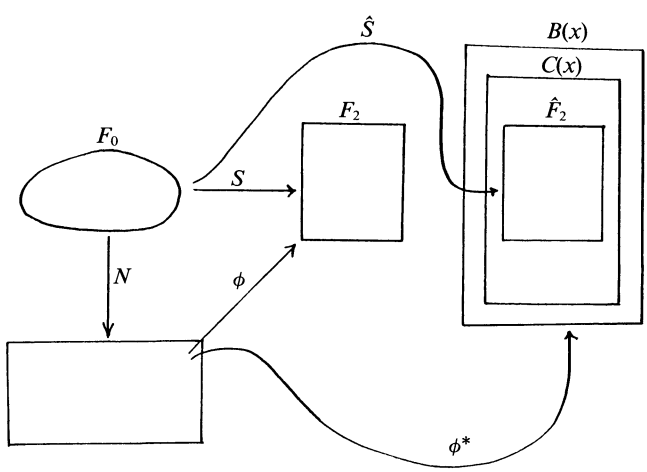


FIG. 6. *Linear optimal algorithm with extended codomain.*

THEOREM 3. *Given a linear problem defined by S: $F_1 \to F_2$ and N: $F_1 \to K^n$ and a convex, balanced subset F_0 of F_1, there exist*:

(i) *A compact Hausdorff space X such that F_2 is isometrically isomorphic to a subspace $\hat{}(F_2)$ of $\mathscr{B}(X)$, the bounded K-valued functions on X with the* sup *norm.*

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

(ii) *A linear optimal algorithm* $\Phi^*: N(F_0) \to \mathscr{B}(X)$ *satisfying*

$$\|\Phi^*(N(f)) - \hat{}(S(f))\| \leqq r(S, N)$$

for all $f \in F_0$, *where* $\hat{}(S(f))$ *denotes the isometric image of* $S(f)$ *in* $\hat{}(F_2)$.

Proof. (i) It is a standard corollary to the Banach–Alaoglu theorem that any normed linear space is isometrically isomorphic to a subspace of $\mathscr{C}(X)$, the continuous K-valued functions on some compact Hausdorff space X. Though we do not need more details for our purposes, it is worth noting that the space X is the unit ball of the conjugate space of F_2 endowed with the weak* topology and the isometric action is provided by the *Gelfand map* $\hat{}$, which imbeds F_2 in its second conjugate F_2^{**}. (See Packel [8] (or most other introductory functional analysis texts) for details.) We shall, in fact, need the larger space $\mathscr{B}(X)$ to hold our embedding.

(ii) For each fixed $x \in X$, consider the linear problem with codomain K defined by N and $S_x(f) = \hat{}(S(f))(x)$. By results 1 and 2 (see Table 1), there exists a linear optimal algorithm $\Phi_x^*: N(F_0) \to K$ such that

$$\|\Phi_x^*(N(f)) - S_x(f)\| \leqq r(S_x, N) \leqq r(S, N) \quad \text{for all } f \in F_0. \tag{1}$$

Letting x vary over X, we must now show that the linear operator Φ^* thus defined on $N(F_0)$ has its range in $\mathscr{B}(X)$. First note that

$$|S_x(f)| = |\hat{}(S(f))(x)| \leqq \|\hat{}(S(f))\| = \|S(f)\|, \tag{2}$$

where the inequality results from $\|\hat{}(S(f))\| = \sup\{|\hat{}(S(f))(x)|: x \in X\}$. Using (1) and (2), we have for all $f \in F_0$,

$$\begin{aligned} |\Phi^*(N(f))(x)| &\leqq |\Phi^*(N(f))(x) - S_x(f)| + |S_x(f)| \\ &\leqq |\Phi_x^*(N(f)) - S_x(f)| + |S_x(f)| \leqq r(S, N) + \|S(f)\|. \end{aligned}$$

Since the final expression is independent of x, $\Phi^*(N(f)) \in \mathscr{B}(X)$ and the proof is complete.

It is tempting to take the view, based on this result, that linear problems do indeed have linear optimal algorithms. We must simply be willing to give the solution operator (and its approximating algorithms) a codomain (namely $\mathscr{B}(X)$) which extends beyond its range. There are problems with this view, however. The extended codomain is generally vastly larger and more complicated than the original. In addition, the members of $\mathscr{B}(X)$ (other than isometric images from F_2) may have no real or abstract connection with the members of F_2. This leads to difficulties in relating a linear optimal algorithm (assuming we can constructively find one) to the original problem. However, the fact remains that a small but significant reformulation of the standard linear setting allows for linear optimal algorithms in a very general context. It is also possible that linear optimal algorithms might be obtained for less drastic extensions to the codomain of S. We take up this question briefly in the next section.

Finally, result 8 is of considerable interest in terms of both its evolution and its content. The idea behind the basic counterexample emerged from an effort by Werschulz to model the problem of inverting a finite Laplace transform within the information-based framework. He then observed that the resulting linear problem could only have linear algorithms with infinite error. Woźniakowski showed how to construct nonlinear algorithms with finite error for such problems. By abstracting this example Werschulz and Woźniakowski [20] develop a class of linear problems with radii of information arbitrarily close to 0 for which all linear algorithms have infinite error.

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

Thus the positive thrust for the existence of linear optimal algorithms made by result 7 is swiftly and decisively parried by result 8. In addition, result 8 includes the "naturally occurring" problem of inverting a finite Laplace transform (though it should be pointed out that one of the norms used to generate the counterexample is far from natural). Our faith in linear algorithms for linear problems is alternately strengthened and shaken, leaving us in a state of tantalizing mathematical ambiguity.

6. One more (counter)example. We saw in the previous section that result 7 (Theorem 3) has the drawback of requiring a vastly extended codomain for the solution operator in order to ensure a linear optimal algorithm. This section is motivated by the question of whether the extended codomain $\mathscr{B}(X)$ might at least be shrunk to $\mathscr{C}(X)$. We will exhibit an example of a linear problem in $\mathscr{C}[0, 1]$ that has a linear optimal algorithm with codomain $\mathscr{B}[0, 1]$, but not with codomain $\mathscr{C}[0, 1]$. The example will require only minor modifications to the example developed in §4.

As in §4, let $F_1 = F_2 = \mathscr{C}[0, 1]$ with the sup norm and let F_0 be the unit ball of $\mathscr{C}[0, 1]$. Then the information operator is again $N\colon F_0 \to \mathbb{R}$, $N(f) = \int_0^1 f(t)\,dt$. The solution operator is defined by $S\colon F_0 \to \mathscr{C}[0, 1]$, $S(f)(x) = g(x) \int_0^x f(t)\,dt$, where

$$g(x) = \begin{cases} \dfrac{1}{x}, & x \in \left[\dfrac{1}{4}, \dfrac{1}{2}\right], \\[2ex] \dfrac{1}{1-x}, & x \in \left(\dfrac{1}{2}, \dfrac{3}{4}\right], \\[2ex] 4 & \text{otherwise.} \end{cases}$$

Clearly the function g is carefully contrived to make something happen; a review of Figure 4(c) should help to reveal the general intent. Indeed, for $x \in [\frac{1}{4}, \frac{3}{4}]$ the convex balanced sets Y_x will, thanks to the $g(x)$ factor, be "stretched" so that their boundary points on the horizontal axis all coincide, forcing a common radius of information $r_x = 1$ (see Fig. 7). Recall that the slopes of bounding tangent hyperplanes at $(\pm r_x, 0)$ for each fixed x determine linear optimal algorithms for the corresponding real-valued solution operator S_x. We conclude from the way such slopes must "jump" at $x = \frac{1}{2}$ that any linear optimal algorithm for the problem determined by S and N must be discontinuous at $\frac{1}{2}$ and must hence require $\mathscr{B}[0, 1]$ rather than $\mathscr{C}[0, 1]$ for its codomain. A little calculation shows, in fact, that any linear optimal algorithm Φ^* for the problem must satisfy, for any $y \in N(f)$,

$$\Phi^*(y)(x) = \begin{cases} 0, & x \in \left[\dfrac{1}{4}, \dfrac{1}{2}\right), \\[2ex] \dfrac{y}{1-x}, & x \in \left(\dfrac{1}{2}, \dfrac{3}{4}\right]. \end{cases}$$

What have we gained for our efforts? It is tempting to conclude, based upon the example above, that the extended codomain of $\mathscr{B}(X)$ in Theorem 3 cannot generally by shrunk to $\mathscr{C}(X)$. Some thought shows that this conclusion would be premature. Even though the solution operator S has no linear optimal algorithm with values in $\mathscr{C}[0, 1]$, applying Theorem 3 lands us in a larger and more mysterious underlying space X for which algorithm values might conceivably be restricted to $\mathscr{C}(X)$, rather than just $\mathscr{B}(X)$.

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

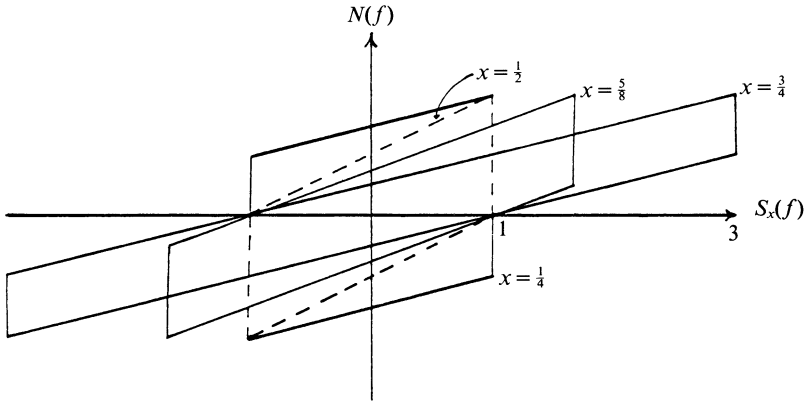


FIG. 7. *Plots of* $\{S_x(f), N(f): f \in F_0\} = Y_x$ *with* $g(x)$ *factor.*

The example, while not serving its initially intended purpose, provides a pleasant surprise. With $\mathscr{C}[0, 1]$ as the codomain of the solution operator (and of any approximating algorithm), the problem determined by S and N has no linear optimal algorithm. Thus we have constructed, from a problem which has at least some degree of naturality, perhaps the simplest counterexample yet found for the nonexistence of linear optimal algorithms for linear problems.

7. Concluding remarks. The setting we have used for discussing error, as embodied in the definition of radius of information and algorithm optimality, has been the traditional "worst-case" setting. Increasingly important in computer science are "average-case" models of error and complexity. By assuming the existence of a probability measure on F_1 (the problem domain), average-case models for information-based complexity can readily be formulated. Work in this area is in its early stages, but a surprising variety of results paralleling the worst-case model have already been obtained (see [4], [17], and [18]). At this time there is no average-case analogue for Theorem 3.

It may have occurred to the reader that we are limiting ourselves by the rigid way we are using information for problem elements. Instead of requiring the n scalar values of information provided by $N(f)$ to be predefined "all at once," it might be advantageous to let earlier values help us decide what information to use for subsequent values. This idea can be formalized by a notion of *adaptive* information as opposed to the nonadaptive information that we have been using. The familiar strategy of finding roots of certain continuous functions defined on a closed interval $[a, b]$ by a bisection algorithm (and the intermediate value theorem) can be formalized by an (adaptive) information-based model. With the adaptive bisection information the radius of information for n function evaluations is essentially $(b-a)/2^n$, dramatically smaller than the radius $(b-a)/(2n)$ attainable with nonadaptive information.

It is important to note that root-finding problems are not linear, so they do not fit the linear framework to which we have restricted ourselves. It is not surprising that adaptively choosing information for such problems should be a valuable strategy. For linear problems with linear information of a given cardinality, it has been shown by Gal and Micchelli [1] and by Traub and Woźniakowski [14] (and this, at first glance, may be a surprise) that there always exists some nonadaptive information that will ensure the minimum possible worst case error. Recent work by Wasilkowski and Woźniakowski [19] has extended this result to certain average case settings.

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

To treat nonlinear problems and others not fitting into our normed linear space framework, a more general approach is needed. The whole theory can be generalized (with some inevitable loss of structure and power) to a surprising degree (see Traub, Wasilkowski, and Woźniakowski [16]).

As the previous paragraphs suggest, information-based complexity has many open problems and research directions. We indicate a few of them here, referring the interested reader to [14] and [21] for a more substantial list. Given a specific solution operator S, a domain F_0, and a cardinality n for the amount of information specified by N, it is often a nontrivial problem to determine the *optimal information* operator N^* for minimizing $r(S, N)$. Even if $r(S, N^*) = \infty$, we can sometimes fix a problem element f and approximate $S(f)$ by a rapidly convergent sequence of algorithm values calculated for information operators with increasing cardinality. This forms the basis for an *asymptotic* setting for defining error, an area that is just beginning to be investigated.

We have focused on theoretical aspects of information-based complexity. There is also a considerable body of literature on applications of the theory to problems involving interpolation, approximation (take the solution operator S to be the identity), differentiation, integration, and differential equations. Work in these traditional areas has confirmed in some cases, and called into question in others, the optimality of classical algorithms while introducing new results and approaches. (An extensive list of references can be found in [14].) Given the generality of the theory and the ubiquitous nature of problems with partial information, potential for a wide variety of additional applications abounds. Some more recent areas of application include convergence of price mechanisms in mathematical economics (where a key result in the economics-directed paper of Saari and Simon [12] is closely related to a theorem in Traub and Woźniakowski [13]), computer vision (Lee [2]), and time series prediction (Milanese, Tempo, and Vicino [6]).

Returning to the basic theoretical question considered in this paper, we have surveyed a variety of positive and negative results on the existence of linear optimal algorithms for linear problems. We hold out some hope for additional and more general positive results. Meanwhile, if pressed for an answer in a few words as to whether linear optimal algorithms exist, we can only respond in our most authoritative voice with a definitive "usually but not always."

REFERENCES

[1] S. Gal and C. A. Micchelli, *Optimal sequential and nonsequential procedures for evaluating a functional*, Appl. Anal., 10 (1980), pp. 105–120.

[2] D. Lee, *Optimal algorithms for image understanding: current status and future plans*, J. Complexity, 1 (1985), pp. 138–146.

[3] C. A. Micchelli, private communication, 1978.

[4] ———, *Orthogonal projections are optimal algorithms*, J. Approx. Theory, 40 (1984), pp. 101–110.

[5] C. A. Micchelli and T. J. Rivlin, *A survey of optimal recovery*, in Optimal Estimation in Approximation Theory, C. A. Micchelli and T. J. Rivlin, eds., Plenum Press, New York, 1977, pp. 1–54.

[6] M. Milanese, R. Tempo, and A. Vicino, *Robust time series prediction by optimal algorithms theory*, Proc. 23rd IEEE Conf. on Decision and Control, Las Vegas, NV, 1984.

[7] K. Yu. Osipenko, *Best approximation of analytic functions from information about their values at a finite number of points*, Math. Notes, 19 (1976), pp. 17–23. (In English.)

[8] E. W. Packel, *Functional Analysis: A Short Course*, Krieger, Huntington, NY, 1980.

[9] ———, *Linear problems (with extended range) have linear optimal algorithms*, Aequationes Math., 31 (1986), pp. 18–25.

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms

[10] E. W. PACKEL AND H. WOŹNIAKOWSKI, *Recent developments in information-based complexity*, Bull. Amer. Math. Soc., 17 (1987), pp. 9–36.

[11] D. G. SAARI AND C. P. SIMON, *Effective price mechanisms*, Econometrica, 46 (1978), pp. 1097–1125.

[12] S. A. SMOLYAK, *On optimal restoration of functions and functionals of them*, Candidate dissertation, Moscow State Univ., Moscow, 1965. (In Russian.)

[13] J. F. TRAUB AND H. WOŹNIAKOWSKI, *Optimal linear information for the solution of nonlinear equations*, in Algorithms and Complexity: New Directions and Recent Results, J. F. Traub, ed., Academic Press, New York, 1976, pp. 103–119.

[14] ———, *A General Theory of Optimal Algorithms*, Academic Press, New York, 1980.

[15] ———, *Information and computation*, in Adv. in Computers, 23 (1984), pp. 35–92.

[16] J. F. TRAUB, G. W. WASILKOWSKI, AND H. WOŹNIAKOWSKI, *Information, Uncertainty, Complexity*, Addison–Wesley, Reading, MA, 1983.

[17] ———, *Average case optimality for linear problems*, J. Theoret. Comput. Sci., 29 (1984), pp. 1–25.

[18] G. W. WASILKOWSKI AND H. WOŹNIAKOWSKI, *Average case optimal algorithms in Hilbert space*, J. Approx. Theory, 47 (1986), pp. 17–25.

[19] G. W. WASILKOWSKI AND H. WOŹNIAKOWSKI, *Can adaption help on the average*? Numer. Math., 44 (1984), pp. 169–190.

[20] A. WERSCHULZ AND H. WOŹNIAKOWSKI, *Are linear algorithms always good for linear problems*?, Aequationes Math., 31 (1986), pp. 202–211.

[21] H. WOŹNIAKOWSKI, *A survey of information-based complexity*, J. Complexity, 1 (1985), pp. 11–44.

This content downloaded from
76.94.197.1 on Sun, 03 Mar 2024 23:00:34 +00:00
All use subject to https://about.jstor.org/terms